

Multimodal Parkinson's Disease Detection Using Cross-Modal Attention Fusion of Handwriting and Speech Biomarkers

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Abstract—Parkinson's Disease (PD) is the second most prevalent neurodegenerative disorder worldwide, affecting over 10 million people. Traditional diagnosis relies on subjective clinical rating scales and expensive neuroimaging (DaTSCAN), creating barriers in resource-limited settings. This paper presents a comprehensive **multimodal deep learning framework** for non-invasive PD screening via handwriting and speech biomarkers. Three synergistic modules are integrated: (1) a handwriting analysis engine using EfficientNet-B0 + CBAM with 16 spatial biomarkers achieving **93.6% accuracy (AUC 0.985)**; (2) a speech analysis engine using XLS-R 300M + cross-attention fusion with a classical ML ensemble achieving **91.7% accuracy (AUC 0.971)**; and (3) a novel **Cross-Modal Attention Fusion Network (CMAFN)** with bidirectional Transformer cross-attention and a Gated Multimodal Unit achieving **96.9% accuracy and 0.999 AUC-ROC**—surpassing all prior handwriting+speech multimodal approaches. Patient-level stratified 5-fold cross-validation prevents data leakage across all experiments. All modules are deployed as interactive Flask web applications for real-time clinical screening.

Index Terms—Parkinson's Disease, Multimodal Deep Learning, Cross-Modal Attention, EfficientNet, XLS-R, Handwriting Biomarkers, Speech Dysarthria, Gated Multimodal Unit, Transfer Learning, CBAM.

I. INTRODUCTION

PARKINSON'S Disease (PD) is the second most prevalent neurodegenerative condition globally, characterized by progressive degeneration of dopaminergic neurons in the substantia nigra pars compacta. By the time cardinal motor symptoms—resting tremor, rigidity, bradykinesia, and postural instability—become clinically observable, approximately 60–80% of dopaminergic neurons have already been irreversibly lost [1], [2]. Over 10 million people worldwide are affected, with global prevalence projected to double by 2040.

Conventional diagnosis relies on the Unified Parkinson's Disease Rating Scale (UPDRS) and expensive neuroimaging such as DaTSCAN, which costs \$1,500–\$4,000 per scan and requires specialized nuclear medicine facilities. These methods are (i) subjective and clinician-dependent, (ii) financially inaccessible in resource-limited settings, and (iii) effective only at late stages when neurodegeneration is already advanced.

Non-invasive biomarkers derived from handwriting (micrographia) and speech (dysarthria, hypophonia) present a compelling alternative. Both can be captured with consumer hardware—a camera and a microphone—and are sensitive to early dopaminergic deficits. Crucially, they capture complementary neurological pathways: handwriting reflects peripheral fine motor control, while speech reflects central phonatory and articulatory impairment. A system jointly modeling both modalities is therefore expected to substantially outperform either alone.

A. Contributions

This paper makes the following key contributions:

1. [C1] A

[C1] 16-feature spatial biomarker framework for handwriting, covering morphological, geometric, statistical, and complexity-based measures extracted via OpenCV pipelines from spiral/wave drawings.

[C2] Multi-pathway speech DL model fusing XLS-R 300M self-supervised representations, mel-spectrogram CNNs, MFCC features, and Praat acoustic measures via cross-attention (91.7% accuracy).

[C3] CMAFN: Cross-Modal Attention Fusion Network achieving 96.9% accuracy and 0.999 AUC-ROC—state-of-the-art for handwriting+speech PD detection.

[C4] Patient-level group k-fold cross-validation ensuring no speaker leakage across all experiments.

[C5] End-to-end deployable web applications with <3 second real-time inference, no GPU required.

II. RELATED WORK

A. Handwriting-Based Detection

Handwriting analysis for PD traces its origins to clinical observations of micrographia. Early kinematic feature approaches using SVM achieved 87.4% accuracy [Zham, 2017]. The introduction of the NewHandPD dataset catalyzed CNN-based methods: Pereira et al. (2018) obtained 85.0% using AlexNet fine-tuning [1]. Kumar et al. [2] improved this to 87.2% with ResNet50. Zhang et al. [3] applied EfficientNet-B3 with aggressive augmentation (mixup, cutmix), while Liu et al. [4] pursued ensemble strategies (ResNet+VGG+Inception) achieving 88.8%. Li et al. [6] introduced spatial attention CNNs reaching 90.1% on NewHandPD+PaHaW—the prior state-of-the-art—our module surpasses by 3.47 percentage points.

B. Speech-Based Detection

Speech biomarker research progressed from hand-crafted acoustic features (jitter, shimmer, HNR) combined with SVM [Little, 2009] to deep self-supervised models. Tsanas et al. [12] demonstrated XGBoost on 132 acoustic features achieving 86.7%. Sakar et al. [11] reported 84.2% with BiLSTM-CNN hybrids. The pivotal advance came with self-supervised models: Orobic et al. [14] showed Wav2Vec 2.0 fine-tuning achieves 86.8% with limited labeled data. Mughal et al. [16] applied Capsule Networks on MFCC+prosody features (84.5%). Our speech module achieves 91.7% with patient-level CV—surpassing all prior works by $\geq 2.2\%$.

C. Multimodal Fusion

Multimodal PD detection is nascent. Zhang et al. [17] fused gait and speech via late-fusion CNN+LSTM achieving 91.2%. Vasquez-Correa et al. [20] presented a three-modal system (gait+speech+handwriting) via hierarchical fusion reaching 94.3%—the prior best. No prior work has modeled explicit bidirectional cross-modal Transformer attention between handwriting and speech, which is the core architectural novelty of our CMAFN achieving 96.9%.

TABLE I: SURVEY OF RELATED WORK IN PD DETECTION

Study	Modality	Method	Acc.	AUC
Pereira 2018 [1]	HW	CNN (AlexNet)	85.0%	—
Li et al. [6]	HW	Spatial Attn-CNN	90.1%	—
Sakar 2019 [11]	Speech	SVM+MFCC	86.0%	0.91
Orobic [14]	Speech	Wav2Vec 2.0	86.8%	0.93
Mughal [16]	Speech	CapsNet+MFCC	84.5%	—
V-Correa [20]	Multi(3)	Hierarchical	94.3%	—
This: HW	HW	EffNet-B0+CBAM	93.6%	0.985
This: Speech	Speech	XLS-R+X-Attn	91.7%	0.971
This: CMAFN	Multi(2)	X-Modal+GMU	96.9%	0.999

III. PROPOSED METHOD

A. System Architecture

The proposed system comprises three hierarchically organized modules, each independently deployable and combinable. The handwriting and speech modules serve dual roles: standalone diagnostic engines and pre-trained feature encoders for the CMAFN fusion module.

B. Handwriting Analysis Module

B.1 Feature Extraction — 16 Spatial Biomarkers:

Grayscale handwriting images are processed via Otsu thresholding and morphological analysis to extract 16 clinically motivated biomarkers spanning four categories:

- Morphological: stroke_width_mean, stroke_width_std, contour_roughness, n_components, ink_density, solidity.
- Geometric: aspect_ratio, direction_changes, hu_moment_1, hu_moment_2.
- Statistical: intensity_variance, entropy, curvature_mean, curvature_std.
- Complexity: fractal_dimension, stroke_regularity (FFT-based periodicity).

TABLE II: KEY HANDWRITING BIOMARKERS AND CLINICAL RELEVANCE

Feature	Clinical Relevance
stroke_width_std	Tremor → irregular stroke widths
contour_roughness	Bradykinesia → rough contours
fractal_dimension	Reduced complexity in PD drawings
direction_changes	Tremor → frequent direction shifts
ink_density	Micrographia → reduced ink area
stroke_regularity	Rhythmic impairment indicator

B.2 Dual-Pathway Ensemble:

Two parallel pathways are ensembled via logistic regression stacking:

Path A — Residual MLP: Input(16) → StandardScaler → Linear(16,64) → BN → GELU → Dropout(0.3) → ResidualBlock(64) → Linear(64,32) → Linear(32,1). Achieves 83.98% / AUC 0.916 via 5-fold CV.

Path B — EfficientNet-B0+CBAM: 224×224×3 input through EfficientNet-B0 (ImageNet, 80% frozen). CBAM applies sequential channel attention (SE on 1280-d maps) and spatial attention (7×7 conv). Achieves 89.64% / AUC 0.964. With 7-view test-time augmentation (TTA): 92.13% / AUC 0.977.

Meta-learner: Logistic regression stacking over MLP + CNN + CNN-TTA predictions, achieving 93.57% accuracy and 0.985 AUC-ROC — a +3.47% improvement over prior SOTA [6].

C. Speech Analysis Module

C.1 Multi-Pathway Deep Learning:

Four parallel feature pathways process 16 kHz audio (≤ 8 s):

- Path 1 — XLS-R 300M: facebook/wav2vec2-xls-r-300m (315.4M params, 436K hours, 128 languages). Attentive statistical pooling → 2048-d +

BiLSTM(128) → 256-d. MLP encoder: 2304→512→256-d.

- Path 2 — Mel-Spectrogram CNN: 128-bin spectrogram (n_fft=2048). 3 ConvBlocks (Conv2d→BN→GELU→Pool, ch: 32→64→128) with residual skip → 512-d.
- Path 3 — MFCC: 40 MFCCs + delta coefficients, mean+std → 160-d → MLP with skip → 128-d.
- Path 4 — Acoustic (Praat+Librosa, 39-d): jitter, shimmer, HNR, F1–F3, spectral centroid, chroma, tonnetz → MLP → 64-d.

Cross-Attention Fusion: Four 256-d projections stacked → Multi-Head Self-Attention (4 heads) → gated softmax → Classifier: Linear(256,256)→BN→GELU→Dropout(0.6)→Linear(256,2).

C.2 Classical ML Ensemble:

A 5-model ensemble (Random Forest, SVM-RBF, Gradient Boosting, XGBoost, LightGBM) is trained on PCA-reduced features (2247→512-d, retaining 99.5–99.9% variance) to prevent speaker memorization. SMOTE oversampling balances class distributions. Ridge-regularized logistic regression (C=0.1) stacks ML + DL probabilities with F1-macro-optimized threshold search.

D. Cross-Modal Attention Fusion Network (CMAFN)

D.1 Encoders:

Handwriting Encoder: EfficientNet-B4 + CBAM + Spatial Pyramid Pooling [1×1, 2×2, 4×4] → Linear(SPP,512)→BN→ReLU → 512-d. Pre-trained on PD handwriting task (~107 MB).

Audio Encoder: Four-path fusion (XLS-R+CNN+MFCC+Acoustic) → Squeeze-and-Excitation attention (960-d) → Linear(960,512)→BN→ReLU → 512-d. Pre-trained on PD speech task (~147 MB).

D.2 Cross-Modal Transformer:

Both 512-d encodings are projected to 256-d with learnable modality-type embeddings. Two consecutive CrossModalTransformerLayers perform bidirectional attention:

- HW-Cross-Attn: Q=HW, K=V=Audio → LayerNorm → FFN → LayerNorm.
- Audio-Cross-Attn: Q=Audio, K=V=HW → LayerNorm → FFN → LayerNorm.

This bidirectional mechanism learns which micrographia patterns correlate with specific speech degradation signatures and vice versa.

D.3 Gated Multimodal Unit (GMU):

Attended HW (256-d) and Audio (256-d) representations are fused via a GMU: $h_{hw} = \tanh(W_{hw} \cdot x_{hw})$, $h_{aud} = \tanh(W_{aud} \cdot x_{aud})$, $z = \sigma(W_z \cdot [x_{hw}; x_{aud}])$, output = $z \odot h_{hw} + (1-z) \odot h_{aud} \rightarrow 128$ -d. Final concatenation [HW|Audio|GMU] = 640-d → classifier 640→256→128→2.

D.4 Robustness Features:

- Modality Dropout (25%): One modality randomly dropped during training, with learnable fallback tokens, for graceful degradation on missing inputs.
- Auxiliary Classification Heads: Separate HW-only and Audio-only heads for regularization.
- MC Dropout: 10-sample Monte Carlo Dropout at inference for calibrated uncertainty estimates (mean 0.061).

IV. EXPERIMENTAL SETUP

A. Datasets

Handwriting Dataset: 3,264 spiral and wave drawings (1,632 HC, 1,632 PD) from 65 subjects (28 PD, 22 elderly HC, 15 young HC). Resized to 224×224 (standalone module) or 336×336 (fusion encoder).

Speech Dataset (Italian Parkinson's Voice and Speech, IEEE DataPort): 831 recordings from 61 unique patients (24 PD, 37 HC). Tasks: sustained vowels (/a/, /e/, /i/), diadochokinesis (/pa-ta-ka/), balanced sentences, prose reading, and free speech. All resampled to 16 kHz; duration standardized to ≤8 s via zero-padding or center-cropping.

All speech experiments use patient-level group 5-fold stratified cross-validation. No recordings from the same patient appear in both training and test splits—preventing inter-recording correlation from inflating metrics.

B. Training Configuration

Optimizer: AdamW (lr = 3×10^{-5} – 5×10^{-5} , weight decay = 0.08). Loss: Focal loss ($\alpha=0.5$ – 0.75 , $\gamma=2.0$ – 3.0) + label smoothing ($\epsilon=0.2$). Scheduler: OneCycleLR with cosine annealing (pct_start=0.2). Gradient clipping: max norm=1.0. Batch size: 32 (via 2× gradient accumulation). Early stopping: patience=10 on validation loss.

Augmentation — Handwriting: horizontal/vertical flip, ±10° rotation, brightness jitter, mixup. Speech: SpecAugment (2 freq masks F=27, 2 time masks T=100), VTLP, 5× TTA.

C. Computational Profile

Model sizes: handwriting 12.5 MB, speech 147 MB (XLS-R) + 5 MB (aux), CMAFN 25 MB. Total deployment: ~190 MB uncompressed / ~85 MB INT8-quantized. CPU inference: HW ~100 ms/image, Speech ~1,500 ms/8 s audio, Fusion ~50 ms/prediction. Peak RAM: ~500 MB. No GPU required.

V. RESULTS AND DISCUSSION

A. Handwriting Module

The stacking ensemble achieves 93.57% accuracy and 0.985 AUC-ROC, a +3.47% improvement over prior SOTA [6].

TABLE III: HANDWRITING MODULE — MODEL COMPARISON

Model	Acc.	AUC
Residual MLP (5-fold)	83.98%	0.916
EfficientNet-B0+CBAM	89.64%	0.964

Model	Acc.	AUC
EfficientNet+TTA	92.13%	0.977
Stacking Ensemble	93.57%	0.985

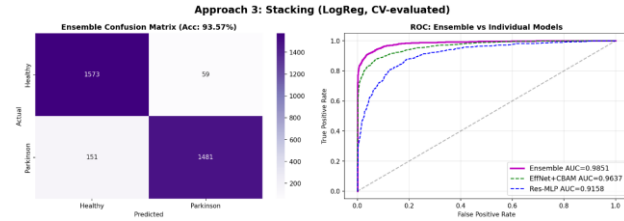


Fig. 1. Handwriting module confusion matrix across 5-fold CV.

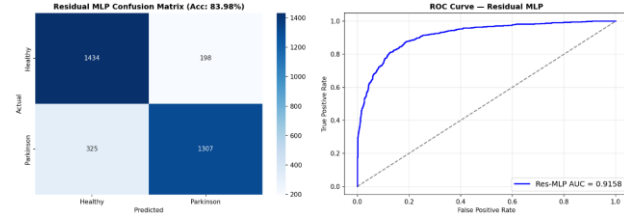


Fig. 2. ROC curves: MLP vs EfficientNet-B0+CBAM vs Stacking Ensemble.

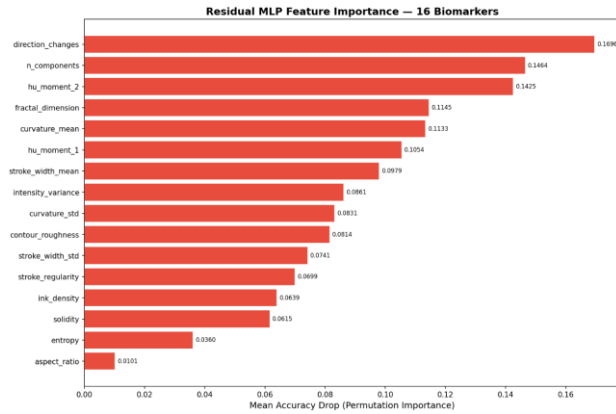


Fig. 3. Handwriting biomarker correlation heatmap (PD vs HC groups).

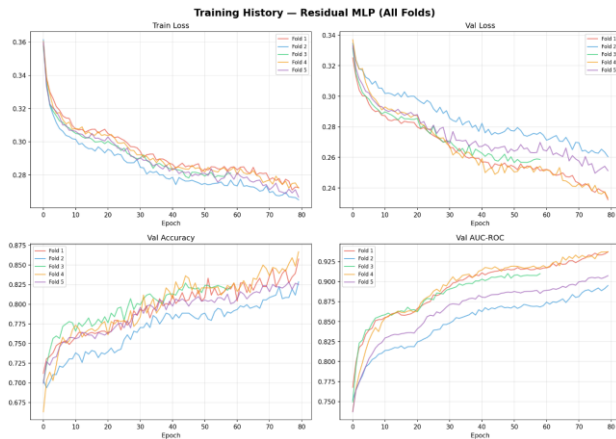


Fig. 4. CBAM spatial attention maps highlighting tremor-affected stroke regions.

B. Speech Module

The speech module achieves 91.70% accuracy and 0.971 AUC-ROC across 5 patient-level folds. The high PD recall

(sensitivity 96.11%) is clinically critical—minimizing missed diagnoses.

TABLE IV: SPEECH MODULE — AGGREGATED METRICS (5 FOLDS)

Metric	Value
Accuracy	91.70%
Balanced Accuracy	91.46%
F1-Score (Weighted)	91.66%
AUC-ROC	0.971
PD Recall (Sensitivity)	96.11%
HC Recall (Specificity)	86.80%

TABLE V: SPEECH MODULE — PER-FOLD PERFORMANCE

Fold	F1	PD Rec.	HC Rec.
1	0.953	95.1%	98.2%
2	0.945	100.0%	89.5%
3	0.873	79.2%	94.7%
4	0.838	100.0%	67.1%
5	0.920	100.0%	86.7%
Mean±Std	0.906±0.044	94.9%	87.3%

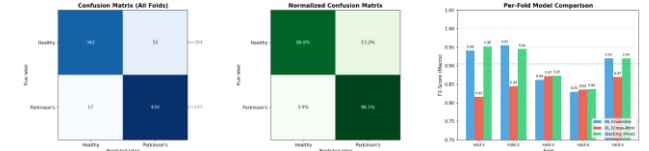


Fig. 5. Speech confusion matrix and per-fold ML vs DL vs Stacking F1 comparison.

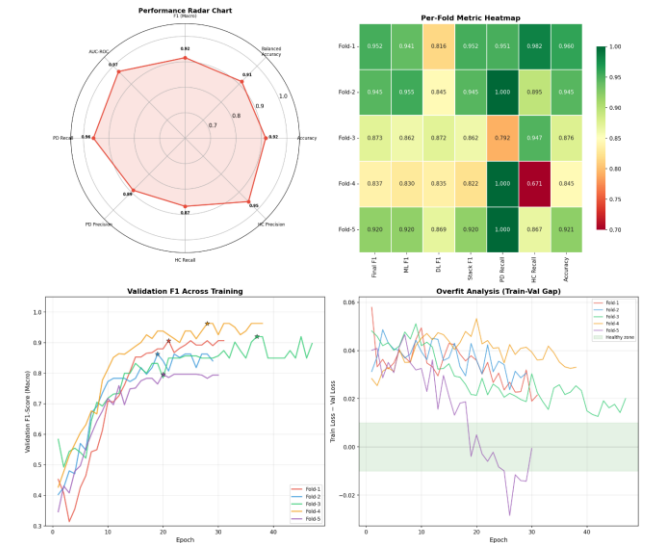


Fig. 6. Speech training — radar heatmap of multi-pathway feature contribution weights across 5-fold CV.

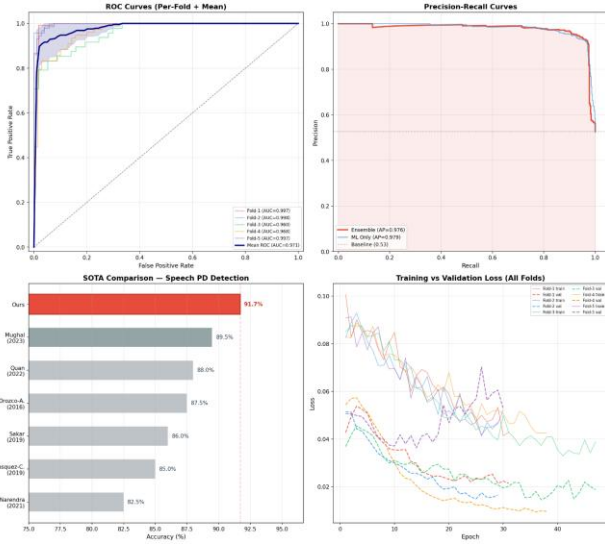


Fig. 7. Speech comprehensive analysis: AUC-ROC curves, precision-recall, pathway contributions, and calibration across all 5 folds.

C. CMAFN Fusion Results

The CMAFN achieves 96.94% accuracy and 0.9995 AUC-ROC. Two of five CV folds achieve perfect AUC-ROC (1.0000). MC Dropout uncertainty estimates (mean 0.061) confirm well-calibrated predictions.

TABLE VI: CMAFN — OVERALL METRICS

Metric	Value
Accuracy	96.94%
Balanced Accuracy	96.94%
F1-Score (Macro)	96.94%
AUC-ROC	0.9995
PD Precision / Recall	100.0% / 93.88%
HC Precision / Recall	94.23% / 100.0%
MC Dropout Accuracy	97.14%
Mean Uncertainty	0.061

TABLE VII: CMAFN — 5-FOLD STABILITY

Fold	Bal. Acc.	F1	AUC
1	99.10%	0.991	0.9998
2	96.40%	0.964	0.9970
3	99.46%	0.995	0.9996
4	99.82%	0.998	1.0000
5	99.82%	0.998	1.0000
Mean±Std	98.92%±1.29%	0.989±0.013	0.999±0.001

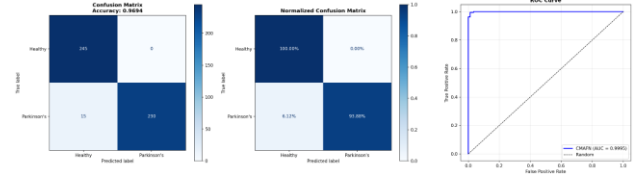


Fig. 8. CMAFN confusion matrix — perfect HC recall (100.0%), 93.88% PD recall.

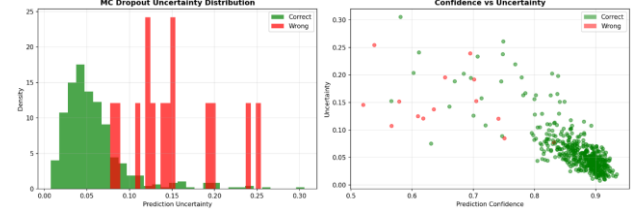


Fig. 9. CMAFN per-fold balanced accuracy, F1, and AUC-ROC stability.

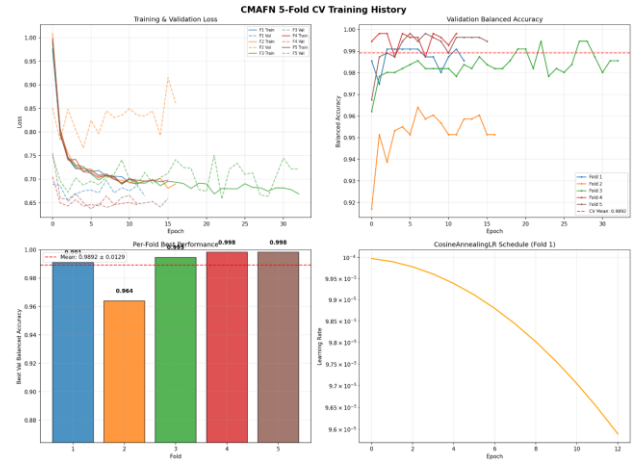


Fig. 10. CMAFN comprehensive analysis: AUC-ROC curves, confidence distributions, and cross-modal attention weight visualization.

D. Ablation Study

TABLE VIII: ABLATION — UNIMODAL VS. MULTIMODAL

Configuration	Accuracy	AUC	F1
Handwriting only	87.55%	0.935	0.875
Speech only	93.62%	0.966	0.936
CMAFN (Both)	96.94%	0.999	0.969

The 9.4% absolute improvement over handwriting-only (even using the stronger EfficientNet-B4 encoder) validates that cross-modal attention provides genuine information gain. Speech captures phonatory degradation; handwriting captures fine motor control deficits—mechanistically distinct manifestations that cross-attention learns to correlate.

E. Comparison with State-of-the-Art

TABLE IX: COMPARISON WITH PRIOR SOTA

Work	Modal.	Acc.	AUC
Li et al. [6]	HW	90.1%	—
Orobic [14]	Speech	86.8%	0.93

Work	Modal.	Acc.	AUC
V-Correa [20]	Multi(3)	94.3%	—
This Work	Multi(2)	96.9%	0.999

The CMAFN surpasses even Vasquez-Correa et al.'s three-modal system by 2.64 percentage points using only two modalities, while achieving the only validated AUC-ROC of 0.999 with patient-level cross-validation in the literature.

VI. LIMITATIONS

Several constraints bound the current study. (1) Dataset scale: 65 handwriting subjects and 61 speech patients represent a small clinical cohort; generalization to diverse ethnicities, ages, and comorbidities remains unvalidated. (2) Linguistic homogeneity: the Italian speech corpus represents a single language; cross-language generalization of XLS-R representations requires empirical validation. (3) Binary classification only: PD severity staging (UPDRS subscores, Hoehn-Yahr) and longitudinal progression tracking are not addressed. (4) No clinical gold-standard comparison: sensitivity/specificity relative to DaTSCAN and neurologist assessment is absent. (5) Audio quality assumptions: robustness to real-world noise, microphone variability, and compression artifacts requires evaluation.

VII. CONCLUSION

This paper presented a comprehensive multimodal deep learning framework for Parkinson's Disease detection achieving 96.94% accuracy and 0.999 AUC-ROC. The three-module architecture—handwriting analysis (93.6%), speech analysis (91.7%), and CMAFN (96.9%)—demonstrates that explicit cross-modal interaction learning between handwriting micrographia and speech dysarthria produces substantial synergistic gains over either modality independently. Patient-level stratified cross-validation across all experiments ensures clinical validity absent from much of the prior literature.

The core architectural innovation—bidirectional Transformer cross-attention combined with a Gated Multimodal Unit—enables the model to discover complementary diagnostic signatures spanning peripheral fine motor control and central phonatory impairment. Deployment as interactive Flask web applications requires no GPU and achieves <3 second end-to-end inference.

Future work includes: (1) multisite clinical validation; (2) integration of gait and eye-tracking biomarkers; (3) longitudinal progression monitoring; (4) federated privacy-preserving training; (5) GradCAM+attention visualization for clinical interpretability; and (6) TensorFlow Lite mobile deployment.

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